



Computational study on catalytic performance of BC₃ and NC₃ nanosheets as cathode electrocatalysts for nonaqueous Li–O₂ batteries

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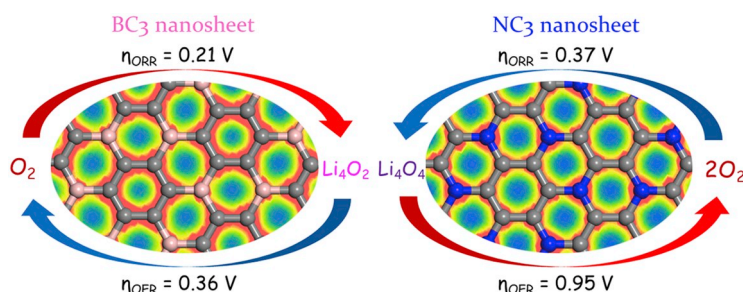
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HIGHLIGHTS

- 2D BC₃ and NC₃ are theoretically explored as cathode catalysts in Li–O₂ battery.
- The ORR pathway and discharge product vary with the cathode catalysts.
- Both the η_{ORR} (0.21 V) and η_{OER} (0.36 V) on BC₃ are considerably low.
- The η_{ORR} (0.37 V) and η_{OER} (0.95 V) on NC₃ are comparable to N-doped graphene.

GRAPHICAL ABSTRACT



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ABSTRACT

The development of high-performance cathode electrocatalysts in nonaqueous lithium-oxygen (Li–O₂) battery is mainly restricted by high charge/discharge overpotentials. Based on first-principles calculations, we theoretically explore the application potential of BC₃ and NC₃ nanosheets as the suitable cathode electrocatalysts. By analyzing the stable adsorption configurations of Li_xO_{2y} (x = 0–4, y = 0–2) intermediates on BC₃/NC₃, it is confirmed that oxygen reduction reaction (ORR) on BC₃ follows four-electron pathway with (Li₂O)₂ as the discharge product, while for ORR on NC₃, (Li₂O)₂ is the discharge product following two-electron pathway. We then plot the corresponding free energy diagrams of charge/discharge processes and calculate the theoretical discharge and charge overpotentials (η_{ORR} and η_{OER}) along the optimal reaction pathways. Our results show that both the η_{ORR} and η_{OER} on BC₃ are considerably low (0.21 V and 0.36 V), indicating that BC₃ can serve as a high-performance cathode electrocatalyst. Meanwhile, the η_{ORR} and η_{OER} on NC₃ (0.37 V and 0.95 V) are comparable to previously reported materials following two-electron pathway (such as N-doped graphene), exhibiting practical values. Both the BC₃ and NC₃ nanosheets are stable enough for application in Li–O₂ battery, and dimethyl sulfoxide (DMSO) isn't recommended as an organic electrolyte when BC₃ works as the cathode catalyst.

1. Introduction

The development of battery technology is becoming increasingly

important for keeping pace with the electronic market and realizing a more sustainable modern society [1]. Although rechargeable lithium ion batteries (LIBs) based on intercalation mechanisms have achieved

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success in the past few decades and been widely used in daily life, their further application in transportation is hampered by limited energy density [2–4], which leads to an upsurge in the research of metal-air batteries [5]. Among various novel battery systems, nonaqueous lithium-oxygen (Li–O₂) battery is a promising alternative for electrical energy storage of electric vehicles. In a typical nonaqueous Li–O₂ battery, O₂ obtained directly from ambient atmosphere serves as the reactant to capture Li ions transferred from anode to cathode mediated by the cathode catalysts [6–8]. Despite their extremely high energy density, several issues that limit battery performance still need to be addressed to make nonaqueous Li–O₂ batteries more viable, including low power density, high overpotential and poor cycling stability [9–13].

To accelerate the practical applications of nonaqueous Li–O₂ batteries, numerous experimental and theoretical studies have been conducted to improve the performance of electrolytes [14–19], membranes [20,21], anodes [22,23] and cathodes [24–32]. Among them, the sluggish kinetics of the oxygen reduction reaction (ORR) and oxygen evolution reaction (OER) occurring during the discharge/charge process on the cathode are considered to be key factors limiting the performance of nonaqueous Li–O₂ batteries [33–35]. As a result, many potential cathode catalysts have been synthesized and characterized, such as noble metals like Pt and Au [36,37], metal oxides [38,39], and carbon-based composites [40–43]. Among them, graphene-based materials have become the research hotspots of air electrodes for Li–O₂ batteries, owing to their superior properties, such as large specific surface area, high porosity and abundant resources [44–48]. As the main ways to modify graphene, defects and doping can effectively improve the performance of pristine graphene, which has been well explained from the view of theoretical simulation [49–57]. Jiang et al. demonstrated that the existence of special defects is beneficial for reducing the overpotential of the nonaqueous Li–O₂ batteries, without promoting the formation of by-product Li₂CO₃ and the decomposition of electrolytes [51]. Ren and co-workers illustrated that among the various doped graphenes (B, N, Al, Si, and P as dopants), doping of B atoms can decrease the oxygen evolution barrier [53,54]. In addition, it has been found that doped N atoms in graphene act as the effective catalytic sites for activating the air cathode ORR of Li–O₂ batteries [55,56]. Until now, the cathode electrocatalysts with lower ORR and OER overpotentials are still highly desirable to achieve high-performance nonaqueous Li–O₂ batteries.

Recently, metal-free XC₃ nanosheets with two-dimensional (2D) graphene-like honeycomb structure are particularly attractive in the field of catalysis, where X represents elements of IIIA–VA group adjacent to C element [58–63]. Unlike the doped graphene, the heteroatoms in XC₃ nanosheets uniformly distribute in the para-site of six-membered ring [62]. The fact that BC₃ and NC₃ nanosheets could be experimentally synthesized [64–67] results in numerous attentions from both scientific investigations and practical applications. Previous theoretical calculations indicated that BC₃ nanosheet and nanotube show great promise as electrode material with large gravimetric capacities in metal ion batteries [58,59,68,69]. Moreover, among the heteroatom-doped graphene, B and N are the most commonly reported ones as cathode catalysts for Li–O₂ batteries [53–56]. However, the catalytic performance of BC₃ or NC₃ nanosheet in nonaqueous Li–O₂ battery has not been explored yet. Herein, based on first-principles calculations, the optimal growth pathways of Li_xO_{2y} (x = 0–4, y = 0–2) intermediates on the surface of BC₃ and NC₃ are discussed, and the theoretical overpotentials are calculated to evaluate the catalytic activity. This work helps to understand the microscopic catalytic mechanism of carbon-based electrocatalysts, and to further design and discover efficient cathode electrocatalysts in nonaqueous Li–O₂ batteries.

2. Computational methods

The spin unrestricted geometry optimizations and energy calculations were performed by the DMol³ module in Materials studio 7.0 [70]. The exchange–correlation term was described by generalized gradient

approximation (GGA) in the form of Perdew–Burke–Ernzerhof (PBE) functional [71]. Grimme's dispersion correction method (DFT-D2) was employed to include long-range interactions [72]. The atomic orbital basis set was specified as the localized double-numerical basis with polarization functions (DNP). The convergence thresholds for energy change, maximum force, and maximum displacement were set to 10^{−5} Ha, 0.002 Ha/Å and 0.005 Å, respectively. A thermal smearing of 0.005 Ha was applied to the orbital occupation to speed up convergence. The reciprocal space for wave function integration was sampled using the Monkhorst–Pack method with 3 × 3 × 1 k-points grid for geometry optimization, and denser grid was applied for electronic property calculations.

The unit cells of monolayer XC₃ (X = B and N) with P₆/MMM symmetry were modeled following our previous work [73], the 3 × 3 supercell together with a vacuum slab of 20 Å was adopted in this work. The adsorption energy (E_{ads}) of Li_xO_{2y} species (x = 0–4, y = 0–2) on BC₃ or NC₃ was calculated as

$$E_{\text{ads}}(\text{Li}_x\text{O}_{2y}) = E_{*\text{Li}_x\text{O}_{2y}} - E_* - (xE_{\text{Li}} + yE_{\text{O}_2}) \quad (1)$$

where E_{*Li_xO_{2y}} is the total energy for the adsorption system of Li_xO_{2y} adsorbed XC₃ nanosheet, E_{*} is the total energy of clean XC₃ nanosheet, E_{Li} is the energy of Li atom in bulk phase, and E_{O₂} is the energy of O₂ molecule in gas phase (triplet as the ground state).

The consecutive adsorption energy (ΔE_{ads}) of a newly added Li or O₂ in *Li_xO_{2y} was defined as

$$\Delta E_{\text{ads}}(\text{Li}) = E_{*\text{Li}_x\text{O}_{2y}} - E_{\text{Li}} - E_{*\text{Li}_{x-1}\text{O}_{2y}} \quad (2)$$

$$\Delta E_{\text{ads}}(\text{O}_2) = E_{*\text{Li}_x\text{O}_{2y}} - E_{\text{O}_2} - E_{*\text{Li}_x\text{O}_{2(y-1)}} \quad (3)$$

The electrochemical calculations were implemented in the computational lithium electrode (CLE) model [74,75]. The Gibbs free energy change (ΔG) for all elemental steps was calculated as

$$\Delta G = \Delta E + \Delta \text{ZPE} - T\Delta S + neU \quad (4)$$

where ΔE is the change of DFT calculated total energy, ΔZPE is the change of zero-point vibrational energy, T is the temperature (T = 298.15 K in this work), ΔS is the entropy change, and neU is the free energy contribution of applied electrode potential U on proton-electron pairs. The term ZPE and S (only vibrational entropy is considered) can be obtained by calculating the frequency according to the following formulas [75],

$$\text{ZPE} = \sum_i \frac{1}{2} \hbar \nu_i \quad (5)$$

$$S_{\text{vib}} = R \sum_i \left(\frac{\hbar \nu_i}{k_B T} \left(\exp\left(\frac{\hbar \nu_i}{k_B T}\right) - 1 \right) - \ln \left(1 - \exp\left(\frac{-\hbar \nu_i}{k_B T}\right) \right) \right) \quad (6)$$

The Bader charge analysis developed by Henkleman group [76] is applied for quantitative charge analysis by using the Vienna Ab initio Simulation Package (VASP) [77,78], due to its more reasonable results on positive charge assigned on Li. More details on VASP calculations, as well as the comparison on charge population and frequency calculations can be found in Supplementary materials.

3. Results and discussion

3.1. Adsorption of reaction intermediates

The catalytic process on the cathode of nonaqueous Li–O₂ battery involves the transfer of multiple electrons, so it is crucial to clearly understand the electronic properties of catalytic substrates. The slices of electron density of BC₃ and NC₃ are shown in Fig. 1, where red color represents the electron accumulation regions. Since the

electronegativity follows the sequence of $N > C > B$, it is clear that electrons tend to accumulate around carbon rings in BC_3 (Fig. 1(a)) and N atoms in NC_3 (Fig. 1(b)). Unlike graphene, BC_3 and NC_3 are p-type and n-type semiconductors [62], with indirect band gap of about 1.8 eV and 1.0 eV (obtained by hybrid DFT functional) [63,79,80], respectively. The different electronic properties may result in different adsorption selectivity, reaction mechanism and final catalytic performance. Therefore, it is interesting to find the correlation between the electronic properties of these two materials and their catalytic performance as cathode electrocatalysts for nonaqueous $Li-O_2$ batteries.

The efficient adsorption of the reaction intermediates is a prerequisite for the catalytic reaction. Similar to our previous theoretical studies on the siligraphenes [73], the adsorption configurations and adsorption performance of Li and O_2 on BC_3 and NC_3 were considered firstly. The different adsorption sites are marked in Fig. S1 of Supplementary materials, and the corresponding E_{ads} are listed in Table S2. It was computationally confirmed that Li atom prefers to adsorb on hollow site of the carbon ring in BC_3 and NC_3 , with E_{ads} of -1.73 eV and -0.25 eV for the most stable adsorption configuration. O_2 could weakly adsorb on the para site of the B atoms in BC_3 with E_{ads} of -0.14 eV, while repulsively binds to NC_3 with a positive E_{ads} (0.15 eV). Thus for both BC_3 and NC_3 , the binding of Li is the beginning step during ORR catalysis. In addition, to understand the growth mechanism of the discharge products on the cathode of $Li-O_2$ batteries, the initial growth and nucleation pathways were then simulated. We modeled the possible Li-O intermediates ($*Li_xO_{2y}$) involved in the initial nucleation process on the surface of BC_3 and NC_3 , and calculated the E_{ads} of the most stable configurations for $*Li_xO_{2y}$, as listed in Table 1.

For the discharge process on BC_3 , the most stable adsorption configurations of possible Li-O intermediates are displayed in Fig. 2, and the ΔE_{ads} of a newly added Li or O_2 on the adsorption systems are labeled near the corresponding configurations. As the first Li-O intermediate, $*LiO_2$ shows adsorption configuration with two O atoms parallelly adsorbed on the para-positioned B atoms and Li atom bonded with the two O atoms. Such configuration results from two respects: on one hand, the binding of Li could greatly enhance the binding strength of O_2 on the B-B bridge site by significantly increasing the negative charge assigned on O atoms (see Table 1); on the other hand, the adsorption of Li on the adsorbed O_2 ($\Delta E_{ads} = -3.02$ eV) is much stronger than its adsorption on clean BC_3 (-1.73 eV). As a comparison, the adsorption of $*LiO_2$ with Li as the anchoring site only has a much weaker E_{ads} of -2.19 eV (see Fig. S2a). In the subsequent steps, the $*Li_2O_2$, $*Li_3O_2$ and $*Li_4O_2$ are confirmed as stable Li-O intermediates because their calculated $|\Delta E_{ads}|$ of Li (3.04 eV, 2.60 eV and 2.61 eV) are all larger than the $|E_{ads}|$ of Li (1.73 eV) on clean BC_3 . Oppositely, the formation of $*Li_xO_4$ species is energetically unfavorable according to the calculated ΔE_{ads} of O_2 (0.07 eV, -0.02 eV and -0.10 eV) as well as the small negative charge

Table 1

Adsorption energies (E_{ads}) of the possible Li-O intermediates involved in the initial nucleation process on the surface of BC_3 and NC_3 (the intermediates involving in optimal pathway are shown in bold font), together with Bader charge distributed on the adsorbate Li_xO_{2y} ($Q_{Li_xO_{2y}}$), as well as Bader charge distributed on the corresponding O atoms in Li_xO_{2y} ($Q_{O_{2y}}$).

	Li_xO_{2y}	E_{ads} (eV)	$Q_{Li_xO_{2y}}$ ($ e $)	Q_{O_2} ($ e $) ^a	Q_{O_2} ($ e $) ^b
BC_3	O_2	-0.14	–	-1.00	–
	Li	-1.73	0.89	–	–
	LiO_2	-3.16	-1.50	-2.37	–
	Li_2O_2	-6.20	-0.81	-2.56	–
	Li_3O_2	-8.80	-0.17	-2.78	–
	Li_4O_2	-11.42	0.51	-2.95	–
	Li_2O_4	-6.13	-1.02	-2.46	-0.31
	Li_3O_4	-8.83	-0.49	-2.75	-0.35
	Li_4O_4	-11.51	0.20	-2.88	-0.40
NC_3	O_2	0.15	–	-0.21	–
	Li	-0.25	0.88	–	–
	LiO_2	-1.98	0.05	-0.80	–
	Li_2O_2	-3.98	0.30	-1.42	–
	Li_3O_2	-5.41	0.93	-1.67	–
	Li_4O_2	-6.99	0.40	-2.96	–
	Li_4O_4 (i)	-8.95	0.09	-1.61	-1.66
	Li_2O_4	-5.14	-0.09	-0.89	-0.91
	Li_3O_4	-7.07	0.12	-1.53	-0.91
	Li_4O_4 (ii)	-9.13	0.33	-1.55	-1.55

^a The charge assigned on 1st O_2 during four-electron pathway and two-electron pathway.

^b The charge assigned on 2nd O_2 during two-electron pathway.

obtained by the 2nd O_2 molecule, see Table 1. Therefore the nucleation of $(Li_2O_2)_x$ starting from $*Li_2O_2$ is unfavorable with BC_3 as the substrate. The specific reaction steps of the optimal growth pathway on BC_3 should be a four-electron pathway as indicated by the red arrows in Fig. 2, and the total reaction equation can be expressed as,



where Li_2O acts as the initial discharge product on BC_3 nanosheet. Moreover, as we can see in Table 1, the negative charge assigned on O_2 in $*Li_xO_2$ keeps increasing, indicating that the O atoms obtain electron with the increase of Li, according with the nature of oxygen reduction. Such ORR process is highly similar with that on siligraphenes, with Li_2O as the discharge product [67]. In addition, restricted by the surface distribution of para-positioned B sites, Li_2O_2 on different sites hardly agglomerates to form larger $(Li_2O_2)_x$ species, oppositely, it is plausible for agglomerations of Li_2O on BC_3 nanosheet, as shown in Fig. S3.

For the initial nucleation process on NC_3 nanosheet, the situation is quite different. Due to the repulsive interaction between O_2 and NC_3 , the $*LiO_2$ shows adsorption configuration with two O atoms binding to the as-adsorbed Li atom, as shown in Fig. 3, with a much smaller E_{ads} of -1.98 eV compared with that on BC_3 (-3.16 eV). Here, the O_2 shows stable adsorption on Li and obtains electrons of $-0.80 |e|$, indicating that O_2 has started to be reduced by the adsorbed Li. As a comparison, the adsorption of $*LiO_2$ with O_2 as the anchoring site only has a much weaker E_{ads} of -1.45 eV (see Fig. S2b). For the following ORR steps, there appears obvious nucleation process with the formation of $(Li_2O_2)_2$ cluster. Following $Li_2O_2^*$, the binding of Li atom is more energetically favorable than that of O_2 molecule, resulting in the formation of $Li_3O_2^*$. Then, a second O_2 molecule becomes more energetically favorable than Li atom when binds with $Li_3O_2^*$, leading to the formation of $Li_3O_4^*$. Thus the pathway of $* \rightarrow *LiO_2 \rightarrow *Li_2O_2 \rightarrow *Li_3O_2 \rightarrow *Li_3O_4 \rightarrow *Li_4O_4$ is confirmed as the optimal pathway from the view of energetics, as indicated by the red arrows in Fig. 3. The 2nd O_2 is evidently reduced during discharge process proven by the significant negative charge transferred from the substrate, see Table 1. Thus the specific reaction steps of the optimal growth pathway on NC_3 follow two-electron pathway, and the total reaction equation can be expressed as,

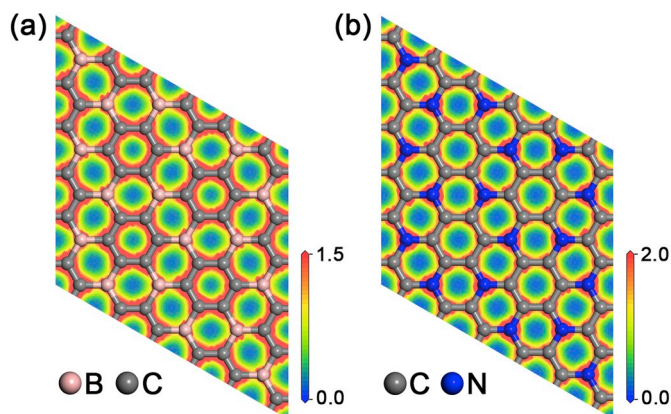


Fig. 1. Slices of deformation electron density of (a) BC_3 nanosheet and (b) NC_3 nanosheet along the nanosheet plane.

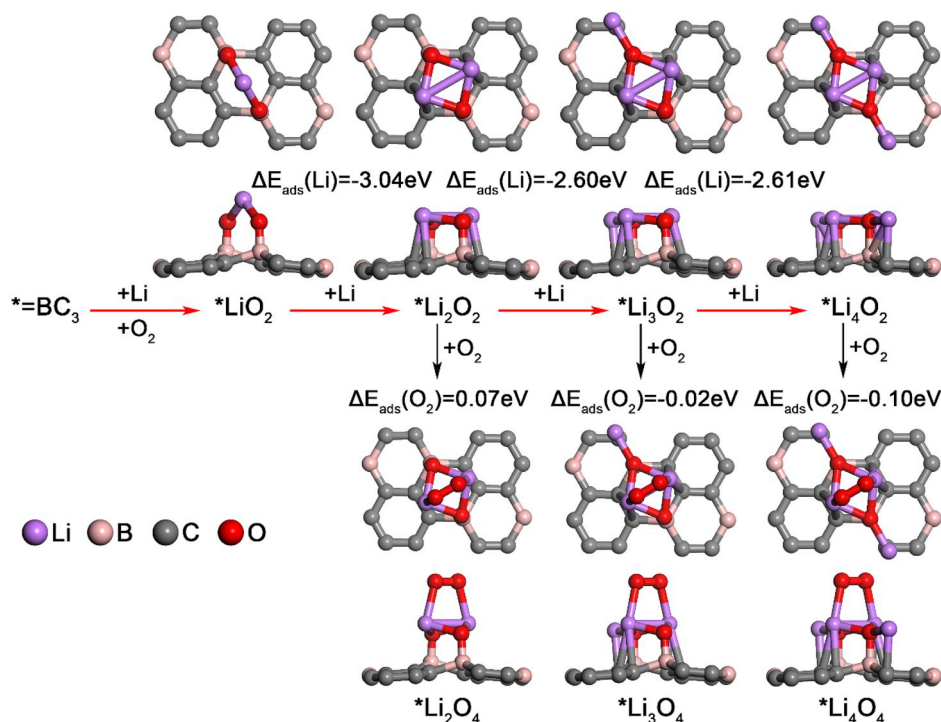


Fig. 2. Top and side views of the most stable adsorption configurations of possible Li_xO_{2y} intermediates involved in the initial nucleation process on BC_3 nanosheet. Red arrows indicate the energetically favorable pathway, while black arrows indicate unfavorable pathways. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)



where Li_2O_2 is the initial discharge product on NC_3 nanosheet. Similarly, the reduction of O_2 molecules can be verified by the increasing negative charge assigned on O_2 (distinguished as 1st and 2nd O_2 in Table 1). Such ORR process is similar with that on pyridinic N-doped graphene, with Li_2O_2 as the discharge product [55,56].

In addition, the density of states (DOS) plots for the $\ast\text{Li}_x\text{O}_{2y}$ along optimal pathways have been depicted in Fig. S4 and Fig. S5. It could be concluded from the DOS plots that during the reaction process, the electrocatalysts keep conductive due to the new peak induced by the adsorption of Li_xO_{2y} intermediates, which ensures the fast discharge/charge with the using of cathode electrocatalysts.

3.2. Discharge/charge process and the overpotentials

The above computational analysis clearly indicates that due to the different doping type, optimal pathways of discharge process on BC_3 and NC_3 are quite different. For ORR on BC_3 , the para-B site exhibits affinity for O_2 binding due to the positive charge assigned on B, similar as the ORR on para-B doped graphene nanoribbon [81]. And the O_2 serves as anchoring site for the following binding of Li, thus limiting the adsorption of a second O_2 as well as the further nucleation of (Li_2O_2) cluster. However, for ORR on NC_3 , the weak affinity against O_2 makes Li be the anchoring site for further binding of O_2 , which is favorable for nucleation of (Li_2O_2) cluster, as shown in Fig. 3.

Based on the optimized structures, we further calculated their free energy changes to elucidate the ORR and OER performance of the XC_3 nanosheets in nonaqueous $\text{Li}-\text{O}_2$ batteries. For a battery, it is well-known that the overpotential value (η) could be the determining criterion to evaluate catalytic activity of the catalyst, which is calculated as the difference between the limiting potential (U_L) and the equilibrium potential (U_{eq}),

$$\eta = |U_L - U_{\text{eq}}| \quad (9)$$

where U_L represents the highest applied electrode potential needed during discharge process (U_{DC}) and the lowest applied electrode potential needed during charge process (U_C). And the theoretical overpotential for the ORR and OER can be further expressed as

$$\eta_{\text{ORR}} = U_{\text{eq}} - U_{\text{DC}} \quad (10)$$

$$\eta_{\text{OER}} = U_C - U_{\text{eq}} \quad (11)$$

The equilibrium potential (U_{eq}) forming Li_xO_{2y} can be obtained from the Nernst equation,

$$U_{\text{eq}} = -\Delta G_f / (ne) \quad (12)$$

where n represents the number of transferred electrons, and the standard formation energy (ΔG_f) was calculated by subtracting the free energy of the isolated Li (xG_{Li}) and O_2 (yG_{O_2}) from the total free energy of Li_xO_{2y} ($G_{\text{Li}_x\text{O}_{2y}}$) by the following formula:

$$\Delta G_f = G_{\text{Li}_x\text{O}_{2y}} - (xG_{\text{Li}} + yG_{\text{O}_2}) \quad (13)$$

In this work, the U_{eq} for four-electron pathway is calculated as 2.86 V, agrees well with the experimental value of 2.91 V [82]. And the U_{eq} for two-electron pathway is calculated as 2.96 V, consistent with the experimental value [82].

The calculated free energy diagram for ORR/OER at zero potential ($U = 0$ V), equilibrium potential (U_{eq}), discharge potential (U_{DC}) and charge potential (U_C) are depicted in Fig. 4. From the free energy diagram we can see that the BC_3 possesses a U_{DC} of 2.65 V and U_C of 3.22 V, while NC_3 possesses a U_{DC} of 2.01 V and U_C of 3.33 V, both well locate at the practical potential range for $\text{Li}-\text{O}_2$ battery measurement (2–4.5 V) [83]. Analyzed from Fig. 4, it is found that for ORR on BC_3 , the potential determining step (PDS) is the formation of $\ast\text{Li}_3\text{O}_2$, due to the lowest energy released from the insertion of the 3rd Li. While for OER on BC_3 , PDS is desorption of $\ast\text{LiO}_2$, resulting from the strongest binding of the LiO_2 group rather than the following Li atoms. For NC_3 , the PDS during ORR is the formation of $\ast\text{Li}_2\text{O}_2$, and during OER it is the decomposition

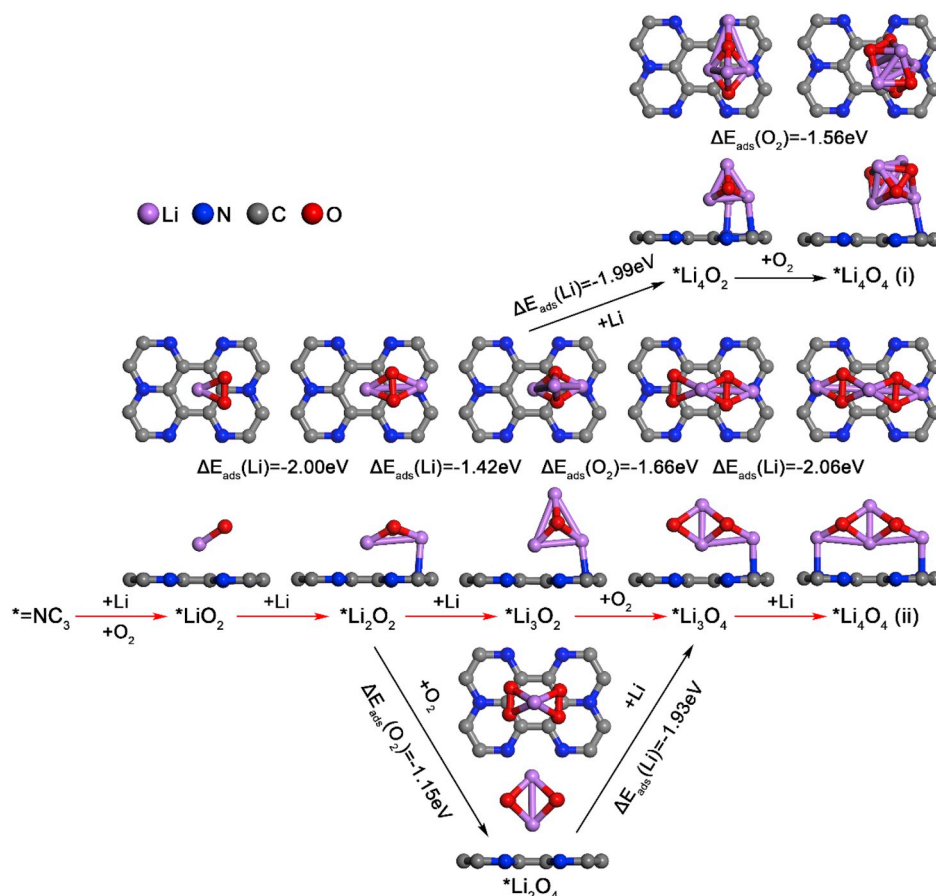


Fig. 3. Top view and side view for the most stable adsorption configurations of possible Li_xO_y intermediates involved in the initial nucleation process on the NC_3 nanosheet. Red arrows indicate the energetically favorable pathway, while black arrows indicate unfavorable pathways. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

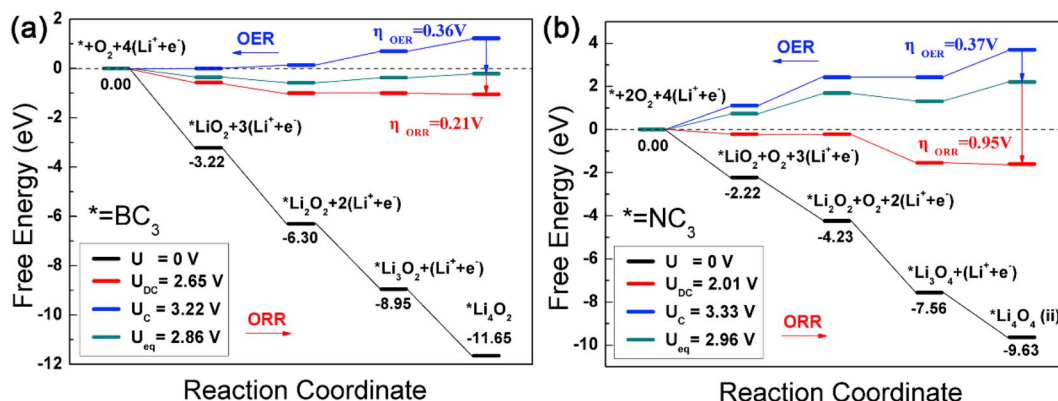


Fig. 4. Free energy diagrams for the charge and discharge process on (a) BC_3 and (b) NC_3 nanosheets.

of $*\text{Li}_3\text{O}_4$.

It is worth noting that the BC_3 exhibits fairly small η_{ORR} (0.21 V) and η_{OER} (0.36 V), as listed in Table 2, both of the values are low enough comparing with the previously reported ones, such as SL-SiC (0.73 V and 1.87 V) [73] or g-GeC (0.89 V and 0.27 V) [84], showing excellent catalytic performance as the cathode electrocatalyst in Li–O₂ battery. The η_{ORR} (0.95 V) of NC_3 are smaller than the previously reported graphitic N-doped graphene (1.06–1.98 V) that sharing the two-electron pathway [55,56], indicating the enhancement of ORR catalytic activity in Li–O₂ battery by doping of N in para-sites. However, both the η_{ORR} and η_{OER} values of NC_3 are still not ideal enough comparing with other

high-performance cathode catalysts for Li–O₂ battery (with Li_2O_2 as the product), such as graphene with DV555-777 defect (0.86 V and 0.03 V) [50]. Therefore, our calculation results indicate that BC_3 could serve as a promising cathode catalyst with the lowest overall overpotential of 0.57 V (the summation of η_{ORR} and η_{OER} values) among the reported ones following four-electron pathway. While NC_3 shows advance comparing with the N-doped graphene, and could also be applicable in nonaqueous Li–O₂ batteries with acceptable overpotential values.

Table 2

The comparison of η_{ORR} , η_{OER} and overall overpotential (unit in V) between BC₃/NC₃ and previously reported cathode catalysts divided by the reaction pathway.

Reaction pathway	Cathode catalyst	η_{ORR}	η_{OER}	Overall overpotential	Ref.
Four-electron	BC ₃	0.21	0.36	0.57	This work
	g-GeC	0.89	0.27	1.16	[76]
	SL-SiC	0.73	1.87	2.60	[67]
Two-electron	NC ₃	0.95	0.37	1.32	This work
	Graphitic N-doped graphene	1.06	–	–	[55]
	Graphitic N-doped graphene	1.56	–	–	[54]
	Graphitic N-doped graphene	1.98	1.60	3.58	[51]
	ML-GeSe	0.94	1.30	2.24	[85]
	Graphene with DV555-777	0.86	0.03	0.89	[50]

3.3. Stability characterization

Except of the high catalytic performance, the stability of cathode catalysts during the usage is also highly related with its practical applications. As reported previously, the structural, thermal and dynamic stability of BC₃ [86] and NC₃ [63] nanosheets have been confirmed by theoretical and experimental investigations. Another issue that should be considered for Li–O₂ batteries with carbon-based cathode catalyst is the formation of Li₂CO₃-like side product, which may lead to a high charge voltage [87]. However, unlike the defective graphene or N-doped graphene that could result in the formation of C–O radical [50,55], the adsorbed O₂ on BC₃ or NC₃ won't directly bind with the C atoms. As shown in Figs. 2 and 3, the O₂ molecule bonded with B atoms in BC₃ and Li above the NC₃ surface. Such indirect binding of O₂ molecule with C atoms could effectively avoid the appearance of C–O radical, thereby suppressing the formation of Li₂CO₃.

Besides, the stability against organic solvent is also an important factor for the cathode catalyst because the electrolyte molecules are in direct contact with the cathode in a practical battery. It is better for the electrolyte molecules to show negligible adsorption on the surface of catalyst to avoid hindering the adsorption of Li or O₂. Dimethyl sulfoxide (DMSO) [88] and 2,3-dimethyl-2,3-dimethoxybutane (DMDMB) [89] have been reported as the stable and applicable organic electrolytes for the nonaqueous Li–O₂ batteries. To check the adsorption performance of the two electrolytes on BC₃ and NC₃, we carried out DFT calculations. The most stable adsorption configuration of DMSO and DMDMB on BC₃ and NC₃ were confirmed and shown in Fig. 5, corresponding E_{ads} (calculated by $E_{\text{ads}} = E_{\text{total}} - E_{*} - E_{\text{electrolyte}}$ and without taking vdW interaction into account here) and transferred charge (Q) are also provided. In addition, the E_{ads} with vdW interactions ($E_{\text{ads(vdW)}}$) and corresponding vdW energies (E_{vdW}) are listed in Table S3.

It is found that there is a significant charge transfer from BC₃ to DMSO, which could be verified by the transferred charge (0.208 e) and the deformation electron density slices as depicted in Fig. 5(a). As a result, the adsorption of DMSO molecule on BC₃ surface exhibits chemisorption characteristics with $|E_{\text{ads}}|$ of 0.44 eV, along with slight structural deformation. As a comparison, the adsorption of DMDMB on BC₃ doesn't induce significant charge transfer, and the $|E_{\text{ads}}|$ (0.24 eV) is also low enough, see Fig. 5(c). Similarly, both the adsorption of DMSO and DMDMB on NC₃ cause negligible charge transfer and the $|E_{\text{ads}}|$ are even smaller than that of DMDMB on BC₃, as shown in Fig. 5(b)&(d). Thus, considering the stability and catalytic efficiency of BC₃ in practical nonaqueous Li–O₂ batteries, DMDMB is recommended as the organic electrolyte in comparison to that of DMSO. While for NC₃ either DMSO or DMDMB is the suitable electrolyte.

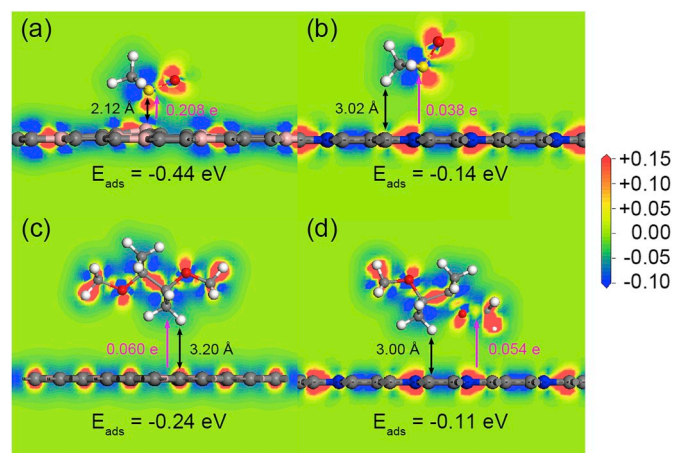


Fig. 5. The deformation electron density slices of the most stable adsorption configuration of (a) DMSO on BC₃, (b) DMSO on NC₃, (c) DMDMB on BC₃, and (d) DMDMB on NC₃. The corresponding E_{ads} , transferred charge (magenta arrow) and nearest distance between the electrolyte molecule and substrate (black arrow) are also marked.

4. Conclusions

Through first-principles calculations, we have theoretically investigated the potential of BC₃ and NC₃ nanosheets as cathode electrocatalysts in nonaqueous Li–O₂ batteries. By analyzing the stable adsorption configurations of Li_xO_{2y} ($x = 0-4$, $y = 0-2$) intermediates on BC₃/NC₃, it is confirmed that ORR on BC₃ follows four-electron pathway with (Li₂O)₂ as the discharge product, while for ORR on NC₃, (Li₂O)₂ is the discharge product following two-electron pathway. The following calculations on discharge/charge potentials obtained by free energy diagrams indicate that both the η_{ORR} and η_{OER} on BC₃ are considerably low (0.21 V and 0.36 V), showing great potential as a high-performance cathode electrocatalyst. Meanwhile, the η_{ORR} and η_{OER} on NC₃ (0.37 V and 0.95 V) are comparable to previously reported N-doped carbon-based materials following two-electron pathway, exhibiting practical values. Calculations on stability indicate that both the BC₃ and NC₃ nanosheets are stable enough for application in Li–O₂ batteries, and DMSO isn't recommended as an organic electrolyte when BC₃ works as cathode catalysts. This work theoretically reveals the great potential of BC₃ as the cathode catalyst in nonaqueous Li–O₂ batteries, and helps to understand the microscopic catalytic mechanism of carbon-based electrocatalysts.

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Appendix A. Supplementary data

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References

- [1] Z. Lyu, Y. Zhou, W. Dai, X. Cui, M. Lai, L. Wang, F. Huo, W. Huang, Z. Hu, W. Chen, *Chem. Soc. Rev.* 46 (2017) 6046–6072.
- [2] J.M. Tarascon, M. Armand, *Issues and challenges facing rechargeable lithium batteries*, in: *Mater. Sust. Energy*, Co-Published with Macmillan Publishers Ltd, UK, 2010, pp. 171–179.

- [3] M.-K. Song, S. Park, F.M. Alamgir, J. Cho, M. Liu, *Math. Sci. Eng. R* 72 (2011) 203–252.
- [4] L. Grande, E. Paillard, J. Hassoun, J.-B. Park, Y.-J. Lee, Y.-K. Sun, S. Passerini, B. Scrosati, *Adv. Mater.* 27 (2015) 784–800.
- [5] F. Cheng, J. Chen, *Chem. Soc. Rev.* 41 (2012) 2172–2192.
- [6] K.G. Gallagher, S. Goebel, T. Greszler, M. Mathias, W. Oelerich, D. Eroglu, V. Srinivasan, *Energy Environ. Sci.* 7 (2014) 1555–1563.
- [7] Z. Chang, J. Xu, X. Zhang, *Adv. Energy Mater.* 7 (2017) 1700875.
- [8] N. Mahne, O. Fontaine, M.O. Thotiyil, M. Wilkening, S.A. Freunberger, *Chem. Sci.* 8 (2017) 6716–6729.
- [9] G. Girishkumar, B. McCloskey, A.C. Luntz, S. Swanson, W. Wilcke, *J. Phys. Chem. Lett.* 1 (2010) 2193–2203.
- [10] R. Padbury, X. Zhang, *J. Power Sources* 196 (2011) 4436–4444.
- [11] M. Park, H. Sun, H. Lee, J. Lee, J. Cho, *Adv. Energy Mater.* 2 (2012) 780–800.
- [12] J. Wang, Y. Li, X. Sun, *Nanomater. Energy* 2 (2013) 443–467.
- [13] P. Tan, H.R. Jiang, X.B. Zhu, L. An, C.Y. Jung, M.C. Wu, L. Shi, W. Shyy, T.S. Zhao, *Appl. Energy* 204 (2017) 780–806.
- [14] Z. Guo, C. Li, J. Liu, Y. Wang, Y. Xia, *Angew. Chem. Int. Ed.* 56 (2017) 7505–7509.
- [15] Y. Qiao, S. Wu, J. Yi, Y. Sun, S. Guo, S. Yang, P. He, H. Zhou, *Angew. Chem.* 129 (2017) 5042–5046.
- [16] J. Han, G. Huang, Y. Ito, X. Guo, T. Fujita, P. Liu, A. Hirata, M. Chen, *Adv. Energy Mater.* 7 (2017) 1601933.
- [17] S. Matsuda, Y. Kubo, K. Uosaki, S. Nakanishi, *J. Phys. Chem. Lett.* 8 (2017) 1142–1146.
- [18] J.-B. Park, S.-H. Lee, H.-G. Jung, D. Aurbach, Y.-K. Sun, *Adv. Mater.* 30 (2018) 1704162.
- [19] B. Tong, J. Huang, Z. Zhou, Z. Peng, *Adv. Mater.* 30 (2018) 1704841.
- [20] U.R. Farooqui, A.L. Ahmad, N.A. Hamid, *Renew. Sustain. Energy Rev.* 77 (2017) 1114–1129.
- [21] W.-J. Kwak, H.-G. Jung, D. Aurbach, Y.-K. Sun, *Adv. Energy Mater.* 7 (2017) 1701232.
- [22] J. Hassoun, H.-G. Jung, D.-J. Lee, J.-B. Park, K. Amine, Y.-K. Sun, B. Scrosati, *Nano Lett.* 12 (2012) 5775–5779.
- [23] H. Song, H. Deng, C. Li, N. Feng, P. He, H. Zhou, *Small Methods* 1 (2017) 1700135.
- [24] H. Cheng, K. Scott, *J. Power Sources* 195 (2010) 1370–1374.
- [25] H.-G. Jung, Y.S. Jeong, J.-B. Park, Y.-K. Sun, B. Scrosati, Y.J. Lee, *ACS Nano* 7 (2013) 3532–3539.
- [26] C. Sun, F. Li, C. Ma, Y. Wang, Y. Ren, W. Yang, Z. Ma, J. Li, Y. Chen, Y. Kim, L. Chen, *J. Mater. Chem.* 2 (2014) 7188–7196.
- [27] J. Zhu, X. Ren, J. Liu, W. Zhang, Z. Wen, *ACS Catal.* 5 (2015) 73–81.
- [28] F. Wu, Y. Xing, L. Li, J. Qian, W. Qu, J. Wen, D. Miller, Y. Ye, R. Chen, K. Amine, J. Lu, *ACS Appl. Mater. Interfaces* 8 (2016) 23635–23645.
- [29] C. Zhao, C. Yu, M.N. Banis, Q. Sun, M. Zhang, X. Li, Y. Liu, Y. Zhao, H. Huang, S. Li, X. Han, B. Xiao, Z. Song, R. Li, J. Qiu, X. Sun, *Nanomater. Energy* 34 (2017) 399–407.
- [30] L. Leng, J. Li, X. Zeng, H. Song, T. Shu, H. Wang, S. Liao, *J. Power Sources* 337 (2017) 173–179.
- [31] Y. Gong, W. Ding, Z. Li, R. Su, X. Zhang, J. Wang, J. Zhou, Z. Wang, Y. Gao, S. Li, P. Guan, Z. Wei, C. Sun, *ACS Catal.* 8 (2018) 4082–4090.
- [32] Y. Hang, C. Zhang, X. Luo, Y. Xie, S. Xin, Y. Li, D. Zhang, J.B. Goodenough, *J. Power Sources* 392 (2018) 15–22.
- [33] Y. Shao, S. Park, J. Xiao, J.-G. Zhang, Y. Wang, J. Liu, *ACS Catal.* 2 (2012) 844–857.
- [34] Y.-C. Lu, B.M. Gallant, D.G. Kwabi, J.R. Harding, R.R. Mitchell, M.S. Whittingham, Y. Shao-Horn, *Energy Environ. Sci.* 6 (2013) 750–768.
- [35] D. Aurbach, B.D. McCloskey, L.F. Nazar, P.G. Bruce, *Nat. Energy* 1 (2016) 16128.
- [36] Y. Xu, W.A. Shelton, *J. Chem. Phys.* 133 (2010), 024703.
- [37] Y.-C. Lu, Z. Xu, H.A. Gasteiger, S. Chen, K. Hamad-Schifferli, Y. Shao-Horn, *J. Am. Chem. Soc.* 132 (2010) 12170–12171.
- [38] T.T. Truong, Y. Liu, Y. Ren, L. Trahey, Y. Sun, *ACS Nano* 6 (2012) 8067–8077.
- [39] J. Zhu, F. Wang, B. Wang, Y. Wang, J. Liu, W. Zhang, Z. Wen, *J. Am. Chem. Soc.* 137 (2015) 13572–13579.
- [40] D. Ji, S. Peng, D. Safanama, H. Yu, L. Li, G. Yang, X. Qin, M. Srinivasan, S. Adams, S. Ramakrishna, *Chem. Mater.* 29 (2017) 1665–1675.
- [41] M. Yu, S. Zhou, Y. Liu, Z. Wang, T. Zhou, J. Zhao, Z. Zhao, J. Qiu, *Sci. China Mater.* 60 (2017) 415–426.
- [42] G. Tan, L. Chong, R. Amine, J. Lu, C. Liu, Y. Yuan, J. Wen, K. He, X. Bi, Y. Guo, H.-H. Wang, R. Shahbazian-Yassar, S. Al Hallaj, D.J. Miller, D. Liu, K. Amine, *Nano Lett.* 17 (2017) 2959–2966.
- [43] Z. Li, S. Ganapathy, Y. Xu, Q. Zhu, W. Chen, I. Kochetkov, C. George, L.F. Nazar, M. Wagemaker, *Adv. Energy Mater.* 8 (2018) 1703513.
- [44] S.-M. Xu, X. Liang, Z.-C. Ren, K.-X. Wang, J.-S. Chen, *Angew. Chem.* 130 (2018) 6941–6945.
- [45] J. Xiao, D. Mei, X. Li, W. Xu, D. Wang, G.L. Graff, W.D. Bennett, Z. Nie, L.V. Saraf, I. A. Aksay, J. Liu, J.-G. Zhang, *Nano Lett.* 11 (2011) 5071–5078.
- [46] Y. Li, J. Wang, X. Li, D. Geng, R. Li, X. Sun, *Chem. Commun.* 47 (2011) 9438–9440.
- [47] B. Sun, B. Wang, D. Su, L. Xiao, H. Ahn, G. Wang, *Carbon* 50 (2012) 727–733.
- [48] X. Zhong, B. Papandrea, Y. Xu, Z. Lin, H. Zhang, Y. Liu, Y. Huang, X. Duan, *Nano Res* 10 (2017) 472–482.
- [49] Y. Li, J. Wang, X. Li, D. Geng, M.N. Banis, R. Li, X. Sun, *Electrochem. Commun.* 18 (2012) 12–15.
- [50] Y. Li, J. Wang, X. Li, D. Geng, M.N. Banis, Y. Tang, D. Wang, R. Li, T.-K. Sham, X. Sun, *J. Mater. Chem.* 22 (2012) 20170–20174.
- [51] H.R. Jiang, P. Tan, M. Liu, Y.K. Zeng, T.S. Zhao, *J. Phys. Chem. C* 120 (2016) 18394–18402.
- [52] H.R. Jiang, T.S. Zhao, L. Shi, P. Tan, L. An, *J. Phys. Chem. C* 120 (2016) 6612–6618.
- [53] X. Ren, J. Zhu, F. Du, J. Liu, W. Zhang, *J. Phys. Chem. C* 118 (2014) 22412–22418.
- [54] X. Ren, B. Wang, J. Zhu, J. Liu, W. Zhang, Z. Wen, *Phys. Chem. Chem. Phys.* 17 (2017) 14605–14612.
- [55] Y. Jing, Z. Zhou, *ACS Catal.* 5 (2015) 4309–4317.
- [56] K.-H. Yun, Y. Hwang, Y.-C. Chung, *J. Power Sources* 277 (2015) 222–227.
- [57] X. Zhang, A. Chen, M. Jiao, Z. Xie, Z. Zhou, *Batteries & Supercaps* 2 (2019) 498–508.
- [58] R.P. Joshi, B. Ozdemir, V. Barone, J.E. Peralta, *J. Phys. Chem. Lett.* 6 (2015) 2728–2732.
- [59] P. Bhauriyal, A. Mahata, B. Pathak, *J. Phys. Chem. C* 121 (2017) 9748–9756.
- [60] X. Li, L. Zhu, Q. Xue, X. Chang, C. Ling, W. Xing, *ACS Appl. Mater. Interfaces* 9 (2017) 31161–31169.
- [61] F. Zheng, H. Dong, Y. Ji, Y. Li, *Appl. Surf. Sci.* 469 (2019) 316–324.
- [62] F. Zheng, H. Dong, Y. Ji, Y. Li, *Int. J. Hydrogen Energy* 44 (2019) 6055–6064.
- [63] H. Zhang, Y. Liao, G. Yang, X. Zhou, *ACS Omega* 3 (2018) 10517–10525.
- [64] K.M. Krishnan, *Appl. Phys. Lett.* 58 (1991) 1857–1859.
- [65] H. Tanaka, Y. Kawamata, H. Simizu, T. Fujita, H. Yanagisawa, S. Otani, C. Oshima, *Solid State Commun.* 136 (2005) 22–25.
- [66] S. Yang, W. Li, C. Ye, G. Wang, H. Tian, C. Zhu, P. He, G. Ding, X. Xie, Y. Liu, Y. Lifshitz, S.-T. Lee, Z. Kang, M. Jiang, *Adv. Mater.* 29 (2017) 1605625.
- [67] H. Yanagisawa, T. Tanaka, Y. Ishida, M. Matsue, E. Rokuta, S. Otani, C. Oshima, *Phys. Rev. Lett.* 93 (2004) 177003.
- [68] J. Zhao, B. Wen, Z. Zhou, Z. Chen, P.v.R. Schleyer, *Chem. Phys. Lett.* 415 (2005) 323–326.
- [69] Z. Zhou, J.J. Zhao, X. Gao, Z. Chen, J. Yan, P.v.R. Schleyer, M. Morinaga, *Chem. Mater.* 17 (2005) 992–1000.
- [70] B. Delley, *J. Chem. Phys.* 113 (2000) 7756–7764.
- [71] J.P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* 77 (1996) 3865–3868.
- [72] S. Grimme, *J. Comput. Chem.* 27 (2006) 1787–1799.
- [73] H. Dong, Y. Ji, T. Hou, Y. Li, *Carbon* 126 (2018) 580–587.
- [74] J.K. Nørskov, J. Rossmeisl, A. Logadottir, L. Lindqvist, J.R. Kitchin, T. Bligaard, H. Jónsson, *J. Mater. Chem. B* 108 (2004) 17886–17892.
- [75] Y. Ji, H. Dong, C. Liu, Y. Li, *J. Mater. Chem.* 6 (2018) 13489–13508.
- [76] G. Henkelman, A. Arnaldsson, H. Jónsson, *Comput. Mater. Sci.* 36 (2006) 354–360.
- [77] G. Kresse, J. Furthmüller, *Phys. Rev. B* 54 (1996) 11169–11186.
- [78] G. Kresse, J. Hafner, *Phys. Rev. B* 48 (1993) 13115–13118.
- [79] X. Tan, P. Jin, Z. Chen, *Phys. Chem. Chem. Phys. : Phys. Chem. Chem. Phys.* 16 (2014) 6002–6011.
- [80] L. Xie, L. Yang, W. Ge, X. Wang, J. Jiang, *Chem. Phys.* 520 (2019) 40–46.
- [81] L. Wang, H. Dong, Z. Guo, L. Zhang, T. Hou, Y. Li, *J. Phys. Chem. C* 120 (2016) 17427–17434.
- [82] M.D. Radin, J.F. Rodriguez, F. Tian, D.J. Siegel, *J. Am. Chem. Soc.* 134 (2011) 1093–1103.
- [83] S.Y. Kim, H.-T. Lee, K.-B. Kim, *Phys. Chem. Chem. Phys.* 15 (2013) 20262–20271.
- [84] Y. Ji, H. Dong, T. Hou, Y. Li, *J. Mater. Chem.* 6 (2018) 2212–2218.
- [85] Y. Ji, H. Dong, M. Yang, T. Hou, Y. Li, *Phys. Chem. Chem. Phys.* 19 (2017) 20457–20462.
- [86] H. Yanagisawa, T. Tanaka, Y. Ishida, E. Rokuta, S. Otani, C. Oshima, *Phys. Rev. B* 73 (2006).
- [87] C. Ling, R. Zhang, K. Takechi, F. Mizuno, *J. Phys. Chem. C* 118 (2014) 26591–26598.
- [88] D. Xu, Z.-l. Wang, J.-j. Xu, L.-l. Zhang, X.-b. Zhang, *Chem. Commun.* 48 (2012) 6948–6950.
- [89] B.D. Adams, R. Black, Z. Williams, R. Fernandes, M. Cuisinier, E.J. Berg, P. Novak, G.K. Murphy, L.F. Nazar, *Adv. Energy Mater.* 5 (2015) 1400867.